

FRB array model

Before and after Dylan's corrections

One combined revision. Before means the work preceding Dylan Gormley's commits. After includes his August archival changes and their September extension.

Numerical boundary: **95d7ed5** (Pranav Sanghavi) to the **complete corrected revision**. The final pre-Dylan commit deleted the manuscript, so paper-before.pdf uses the last retained pre-Dylan manuscript at **c74d910**. That paper has an earlier processing-cost model. Numerical tables and paired Figure 1 panels here use 95d7ed5; see page 6.

Main conclusion

The sampled design preferences survive, but the corrected absolute rates differ substantially. All optimized frequency-doubling ratios remain below 1/4. Dishes still beat aperture tiles at the tested slopes $\alpha = -1.5$ and -2 ; the optimal aperture tile at $\alpha = -1$ remains a single dipole.

Changes in absolute scale

The notebook used a daily Parkes rate under an annual label and an incorrect Boltzmann constant. At a fixed area, frequency and element count, the complete normalization correction gives these multipliers:

alpha	Days per year	Boltzmann factor	Combined factor
-1	365.25	1.32593	484.29489
-1.5	365.25	1.52679	557.66029
-2	365.25	1.75808	642.13975

Changes in designs and curve shapes

Your aperture-cost and budget fixes change element counts, so the full revision does **not** preserve curve shapes. After removing the scale multipliers, optimized aperture rates change by **-12.0% to +14.2%**. Dish peaks change by **-0.4% to +4.3%**. The $\alpha = -2$, 1600 MHz dish optimum moves from the old 29.9 m^2 grid edge to **42.676 m^2** .

Figure 2 and Table 1 also change

Survey uncertainties are passed as error magnitudes, the Parkes 2016 fluence threshold is consistently 4 Jy ms, and its tabulated uncertainties are corrected by a factor of ten. Figure 2 is part of this combined revision; it is not unchanged.

Optimal designs and predicted rates

Before: the actual numbers produced by 95d7ed5, which had incorrect annual labels and budget counts. **After:** annual rates using corrected counts. Before dish optima use the 0.1 m^2 grid ending at 29.9 m^2 ; after dish optima search every affordable integer count. Aperture choices use the same discrete tiles.

Dish arrays

alpha	MHz	Old area	New area	Old n	New n	Old rate	New /yr
-1	400	0.100	0.100	1594	1593	192.32	93,080.8
-1	800	0.100	0.100	836	835	35.66	17,249.9
-1	1600	0.100	0.100	439	438	6.62	3,199.1
-1.5	400	9.200	9.398	1183	1174	593.32	330,609.4
-1.5	800	15.700	15.787	617	615	122.75	68,306.4
-1.5	1600	26.900	26.439	322	323	25.47	14,146.4
-2	400	15.000	15.133	969	964	2,698.03	1,729,891.0
-2	800	25.300	25.422	507	505	622.90	398,753.8
-2	1600	29.900	42.676	310	265	137.61	92,162.7

Aperture tiles

alpha	MHz	Old area	New area	Old n	New n	Old rate	New /yr
-1	400	0.179	0.179	1562	1519	188.46	88,756.9
-1	800	0.045	0.045	827	814	35.28	16,816.0
-1	1600	0.011	0.011	436	432	6.58	3,155.3
-1.5	400	1.609	1.609	1191	1096	250.67	123,399.7
-1.5	800	0.715	0.715	613	587	25.94	13,557.4
-1.5	1600	0.279	0.402	329	292	2.68	1,499.9
-2	400	2.861	2.861	937	879	481.15	271,898.8
-2	800	1.118	1.609	510	426	27.84	17,961.4
-2	1600	0.548	0.715	247	231	1.60	1,173.6

Areas are effective collecting areas in m^2 ; n counts dishes or tiles. At alpha = -1 the dish calculation retains $A \geq 0.1 \text{ m}^2$. That mathematical minimum lies in the dashed small-dish extrapolation and is not a validated practical dish design.

Fixed-tile changes can be larger than changes in the optimum. Across all 117 aperture grid points and three slopes, rates divided by the normalization factors change by 0.84683 to 6.25 times. The 6.25 maximum is at 400 MHz and $m = 1444$ ($A = 258.19 \text{ m}^2$, outside the displayed plot), where the corrected count is 20 instead of 8.

Frequency and feasibility comparisons

Ratio of the optimized rate after doubling frequency

Design	alpha	Frequency change	Before	After
Dish	-1	400 to 800 MHz	0.185427	0.185321
Dish	-1	800 to 1600 MHz	0.185658	0.185457
Aperture	-1	400 to 800 MHz	0.187189	0.189462
Aperture	-1	800 to 1600 MHz	0.186396	0.187635
Dish	-1.5	400 to 800 MHz	0.206880	0.206608
Dish	-1.5	800 to 1600 MHz	0.207489	0.207102
Aperture	-1.5	400 to 800 MHz	0.103501	0.109866
Aperture	-1.5	800 to 1600 MHz	0.103323	0.110635
Dish	-2	400 to 800 MHz	0.230870	0.230508
Dish	-2	800 to 1600 MHz	0.220916	0.231127
Aperture	-2	400 to 800 MHz	0.057862	0.066059
Aperture	-2	800 to 1600 MHz	0.057467	0.065342

The largest absolute change is the dish $\alpha = -2$ ratio for 800 to 1600 MHz: 0.220916 to 0.231127. The aperture ratios change because the collector-cost function and budget counts change. Uniform normalization factors cancel in these ratios.

Effective area at dish diameter $d = 5$ wavelengths

Frequency	Before (m ²)	After (m ²)
400 MHz	22.058771	5.514693
800 MHz	5.514693	1.378673
1600 MHz	1.378673	0.344668

At aperture efficiency 0.5, effective area is $0.5 \pi (2.5 \lambda)^2$. The previous code divided by 0.5. Corrected line-style boundaries are one-quarter the old areas; this plotting correction does not itself change rates.

Figure 1: before and after

Before panels are the images committed at 95d7ed5, using its N log N processing model. After panels are regenerated from the corrected notebook. The x range expands from 0-30 to 0-60 m^2 ; y limits scale by the page 1 factors. Shapes also change because aperture and budget counts are corrected.

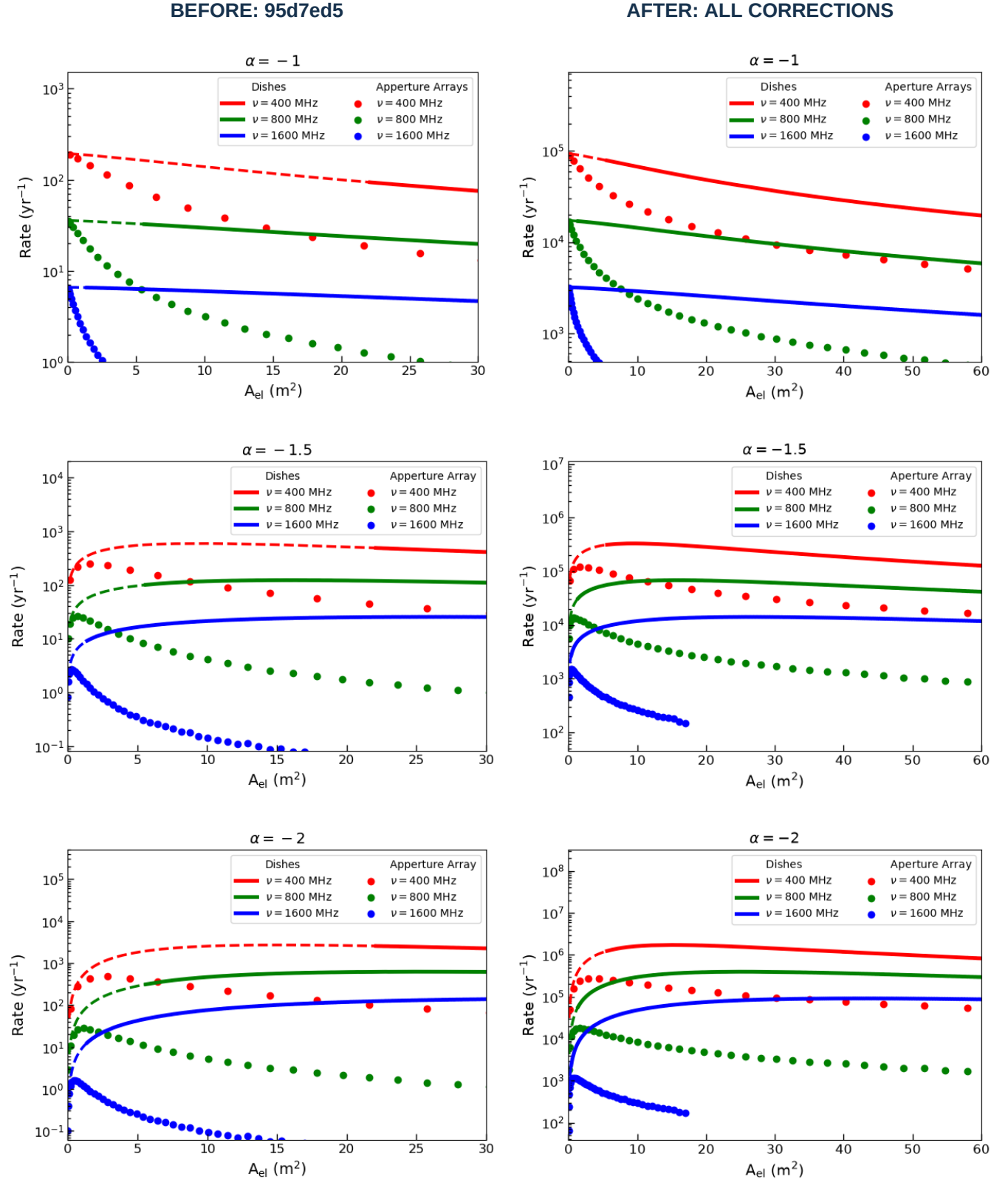
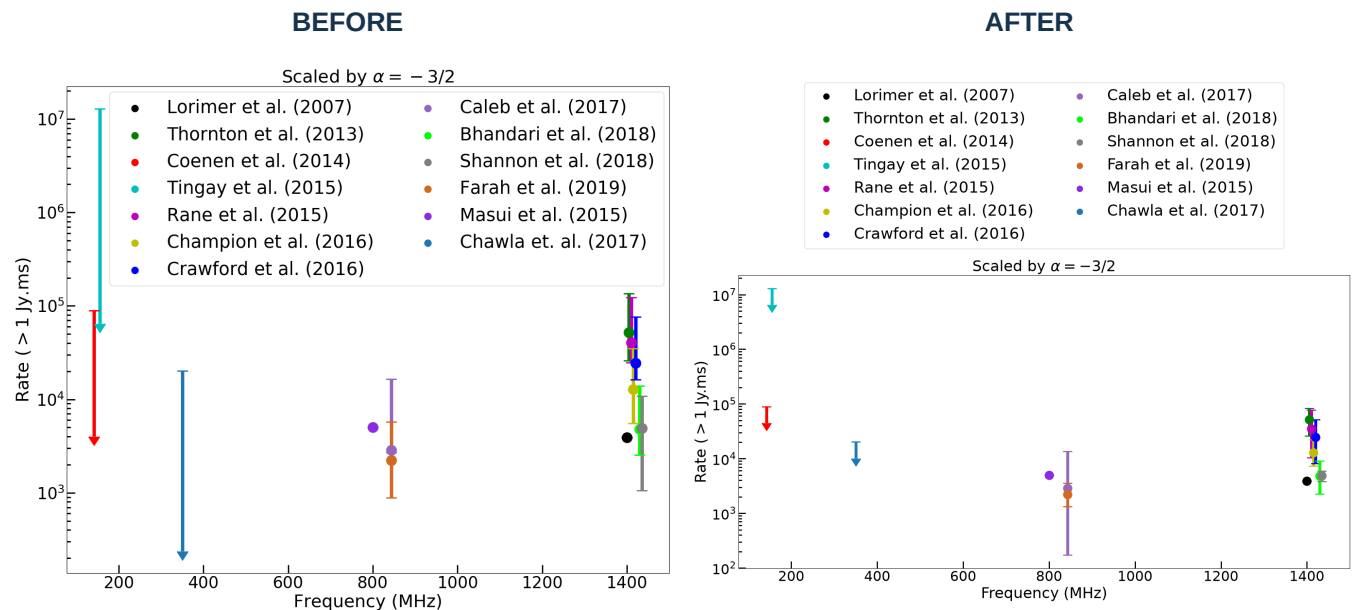


Figure 2: survey rates and uncertainty

Both panels scale the cited surveys to a common 1 Jy ms threshold using $\alpha = -3/2$. The historical image comes from c74d910; its survey notebook is byte-identical at 95d7ed5. The after panel includes your August corrections.



Examples of displayed central rates and intervals

Survey	Old center	Old interval	New center	New interval
Thornton	51,962	25,981 - 135,100	51,962	25,981 - 83,138
Rane	40,610	24,800 - 122,820	35,200	10,400 - 76,800

These are scaled survey rates per sky per day, not the annual model rates on page 2. Intervals are those actually drawn from the notebook's central value and error magnitudes.

What the correction does

Error bars: Matplotlib expects distances from the central value. Supplying absolute interval endpoints made the old displayed intervals wrong; the new arrays use separately scaled positive lower and upper uncertainties.

Parkes 2016 / Rane: changing 4.4 to the cited 4 Jy ms lowers the scaled center from 40,610 to 35,200 (13.32%). Its corrected interval is 10,400-76,800. Table 1 now gives the unscaled rate as 4400 with +5200/-3100 uncertainty, replacing +520/-310.

Upper limits: the underlying three limit values are unchanged. Arrow lengths are now proportional to the plotted values; they are presentation choices, not statistical uncertainty intervals. The other survey centers are unchanged. The legend now sits above the axes so every limit marker remains visible.

Provenance and validation

The authorship boundary

Before calculations: 95d7ed56d345e4a4d7bd12fece20bbbf6e56615c, Pranav Sanghavi, 20 May 2020. **After:** Dylan Gormley's single consolidated correction commit, including the August work and its September extension. The notebook source hashes in `comparison-data.json` identify the calculations used here.

Before manuscript: c74d910faa1dffde60868d0f05260b2a8b1cf5f5 (v1.0), the last manuscript retained in the original repository before Dylan's work. It uses a linear processing approximation. The subsequent pre-Dylan notebook uses an explicit $N \log N$ term. The after paper also incorporates a later working draft restored during archiving. Therefore a paper-to-paper textual diff includes that draft history; it cannot attribute every change solely to numerical corrections.

What changes in the corrected manuscript

The abstract's optimum-area range extends to about 43 m^2 . Cost prose records the implemented aperture coefficient and processing terms. Figure 1 and the rate normalization use consistent annual units. Sections 5-7 give corrected yields, optima and frequency ratios. Figure 2 and Table 1 include the survey uncertainty fixes. The after PDF identifies itself as an unpublished working draft.

Verification completed

Six regression tests pass: independent SI-unit rate calculation; all 1,917 modeled counts within budget and maximal; the 5-wavelength geometry; the off-grid optimum against independent numerical roots; sampled frequency/design comparisons; and canonical error magnitudes at all three slopes. Both notebooks execute successfully (20 code cells combined). Both manuscript PDFs have five pages.

The before rebuild changes only TeX compatibility: removes the redundant `amssymb` load and unsupported table placement, and supplies an empty bibliography series field. Historical text, figures and publication placeholders remain. Its legacy equation layout warning is retained. The after manuscript resolves citations and has no overfull boxes. All delivered PDF pages were visually checked.

Scientific scope

The common normalization factors alone are 484-642; these are not small changes in absolute predictions. The qualitative conclusions above concern only the tested frequencies, slopes and cost assumptions. They do not validate the power-law extrapolation or every historical survey measurement, nor establish a preferred design for every continuous value of α .

Sources and reproducibility

Bhandari et al. (2018) (daily normalization); Rane et al. (4 Jy ms and uncertainties); NIST constants; NRAO, equation 84 (aperture efficiency); `Matplotlib errorbar`. JSON and CSV files alongside this report contain the numbers. The comparison directory contains source diffs and build scripts. The corrected working draft and this report are distributed together; the v1.0 tag remains unchanged.